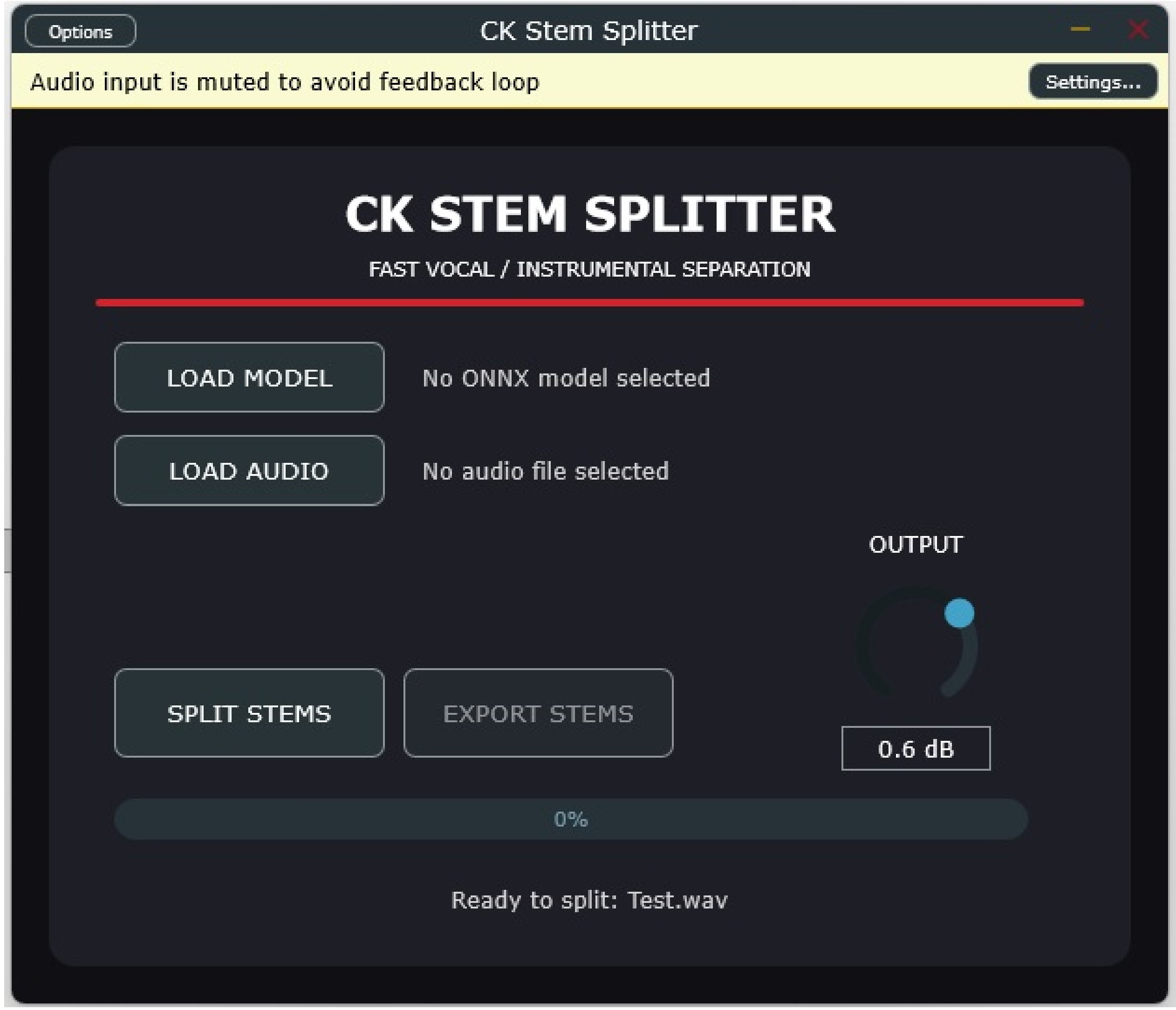
CK STEM SPLITTER 4 STEMS



Tools used:

- Cmake v4.3.1
- Juce v9.1.0
- Visual Studio Community 2026 v18.9.2
- Inno Setup v7.1.0
- Python v3.12.11
- demucs-onnx 0.3.4
- Asio sdk v2.3.4, Vst-sdk v3.8.0-build66, Vst-sdk v2.4, lv2-sdk v1.18.10, aax-sdk v2.8.1, Jack2 v1.9

This fork is a 4-stem version (bass, drums, vocals, other) of Milkstyles' CKStemSplitter (originally a two-stem tool). Various changes have been made to both the code and the Python engine (such as activating the “Shift Tricks” parameters to reduce robotic artifacts).

The “small” version uses the “htdemucs.onnx” model (301 MB), a non-FT StemSplitio model available at the following link: https://huggingface.co/StemSplitio/htdemucs-onnx Installing the “small” version requires approximately 420 MB of free space. This version still allows you to load models other than the default one, provided they are ONNX models designed for 4-stem extraction.

The “large” version uses a different engine, built using a script that enables the use of four models (although loading just one via the app interface is sufficient); these are still StemSplitio models, but in this case, they are fine-tuned and specialized individually for drums, bass, vocals, and other elements available at the link https://huggingface.co/StemSplitio/htdemucs-ft-onnx , Each model is 301 MB in size (not 316 MB as stated on the website), totaling 1.17 GB. Audio separation will take slightly longer but will be more accurate. However, it will not be possible to use other models: although the button for loading different models remains active, the process will either fail to complete or lack accuracy, as the engine is designed to load exactly those four specific models. In fact, to start the process, you simply need to load any one of the models—"htdemucs_ft_bass.onnx", "htdemucs_ft_drums.onnx", "htdemucs_ft_vocals.onnx", or "htdemucs_ft_other.onnx"—and the engine script will automatically load the other models located in the same folder. To use different models, they would need to be built according to the same parameters and specifications used for these ones. The script incorporates—and adapts—the "bag_infer.py" file available on the same Hugging Face page as the models. Using htdemucs-onnx means the end user does not need to install Python or other tools on their PC.

The two versions cannot be installed together because they would overwrite each other; furthermore, they use the same name and the same Ids. The installer for both versions places its components in the following directories:

- the plugins in their default folders:
  - "C:\Program Files\Common Files\VST3\CK Stem Splitter.vst3"
  - "C:\Program Files\Common Files\VST2\CK Stem Splitter.dll"
  - "C:\Program Files\Common Files\LV2\CK Stem Splitter.lv2"
  - "C:\Program Files\Common Files\Avid\Audio\Plug-Ins\CK Stem Splitter.aaxplugin"

- the model in:
  - "C:\ProgramData\Commercial Kings\CK Stem Splitter\engine\models"

- the engine in:
  - "C:\ProgramData\Commercial Kings\CK Stem Splitter\engine"

the cache folder being used is:

- “C:\Users\Eugenio\AppData\Roaming\Commercial Kings\CK Stem Splitter\Cache”

USING THE STANDALONE VERSION

There is no need to open your DAW; to save time, simply launch your standalone app (CK Stem Splitter) directly:

- Click LOAD AUDIO and choose a test track.
- Click LOAD MODEL, navigate to your "Modelli_Test" folder, and select any of the 4 ONNX files (for example, select htdemucs_ft_vocals.onnx). Remember: the new Python script will use that file's folder to automatically locate the other three.
- Press SPLIT / ANALYZE.

You will see it processing the track (this will take some time, as the 4 models combined are resource-intensive and it is processing 4 high-quality files). Once finished, the bar will reach 100%, the EXPORT STEMS button will light up, and you can save the perfect tracks!

Saving the 4 tracks involves two distinct phases: an automatic process running "behind the scenes" and a manual step selected by the user.
When the engine finishes analyzing a track (at approximately 88% on the progress bar), it generates four .wav files: `drums.wav`, `bass.wav`, `other.wav`, and `vocals.wav`. These files are saved to the cache folder. Once extraction is complete, the "Export Stems" button becomes active; clicking it opens the standard Windows dialog box, allowing the user to select a destination folder, after which the four .wav files are copied from the cache folder to that chosen location.

Extraction is always performed in stereo format.

USING PLUGINS

The plugin sends the tracks to the DAW. Routing is handled within the DAW: you create four audio tracks in your project and assign their inputs to the plugin's four outputs. The gain control on the plugin's GUI affects all four tracks simultaneously; individual track volume adjustments, as well as Solo and Mute functions and effect slots (e.g., applying reverb only to the vocal track), are managed directly on the DAW tracks.

The sample rate is always determined by the host application (Ableton, Cubase, Reaper, etc.) rather than the plugin itself; the plugin adapts to the user's project. When the DAW loads the plugin, it communicates the current sample rate (e.g., 44.1 kHz, 48 kHz, 96 kHz). Although Demucs models natively extract audio at 44.1 kHz, the plugin performs real-time resampling in the background—leveraging JUCE's `AudioFormatReaderSource` and `AudioTransportSource` classes—so it will operate at whatever sample rate the user has set in the DAW.

DAW SETUP

Ableton Live

In Ableton, the trick is to load the VST onto a "Master/Source" track and then create 4 Audio tracks that pick up the specific signal from that plugin.

Source Track:
Insert the CKStemSplitter plugin onto an empty track (Audio or MIDI). Rename it, for example, [STEM ENGINE].
Create 4 Audio Tracks:
Insert 4 new Audio tracks into the project and rename them Vocals, Bass, Drums, and Other.

Configure the input (Audio From) for each track:
Vocals Track:
Under Audio From, select the [STEM ENGINE] track.
In the drop-down menu immediately below, select 1/2-CKStemSplitter (or 1/2-Vocals).
Set monitoring to In.
Bass Track:
Audio From -> [STEM ENGINE]
Below -> 3/4-CKStemSplitter
Monitor -> In
Drums Track:
Audio From -> [STEM ENGINE]
Below -> 5/6-CKStemSplitter
Monitor -> In
Other Track:
Audio From -> [STEM ENGINE]
Below -> 7/8-CKStemSplitter
Monitor -> In

Cockos REAPER

REAPER handles multi-output plugins in an extremely versatile way using track channels and the routing matrix (Sends).

Automatic Method (A few clicks)

Open the FX window on the desired track.

Right-click on the CKStemSplitter plugin in the effects list.

Select "Build multichannel routing for output of selected FX...".

REAPER will automatically create 4 separate tracks and set up all the input connections for you.

Manual Method

Load the plugin onto Track 1 (rename it [STEM ENGINE]). Ensure Track 1 is set to 8 channels (if you click the "2 in 8 out" button at the top of the VST window, you will see that channels 1 through 8 are active).

Disable "Master send" on Track 1 (click the ROUTE button on Track 1 and uncheck "Master send") to prevent the summed audio from outputting directly to the master bus.

Create 4 new tracks below Track 1: Vocals, Bass, Drums, Other.

Open the ROUTE window for Track 1 and create 4 Sends:

Send 1 to the Vocals track: Audio 1/2 -> 1/2

Send 2 to the Bass track: Audio 3/4 -> 1/2

Send 3 to the Drums track: Audio 5/6 -> 1/2

Send 4 to the Other track: Audio 7/8 -> 1/2